thickness must never be less than the thickness of the plate to which it is applied.

In case of patches applied prior to July 1, 1914, if there is no authentic record of the date when or the shop where the alteration was made, insert the word "Unknown" in the proper blank spaces. It will not be necessary to report application of patches on surfaces supported entirely by staybolts.

Where old patches have been removed from boilers and replaced by new ones, a complete sketch of the new patch must be shown on Form No. 19, and a notation shown thereon stating that it replaces an old patch.

Where authentic records of the tests of boiler plates used in making repairs or constructing new boilers exist, the lowest tensile strength as covered by test reports must be shown on Form No. 19.

It will also be necessary to indicate on Form No. 19 transfers of all locomotive boilers from one locomotive to another when such changes are made, giving the builder's numbers, or boiler serial number in conjunction with the locomotive number with each and every case.

It is extremely important that all concerned in the repairs of boilers be thoroughly acquainted with the manner in which the application of patches are to be handled, and it must be further understood that three (3) copies of Form No. 19 are to be forwarded to the office of Mechanical Engineer as soon as possible, after such changes have been made, as they must be filed with the Interstate Commerce Commission within thirty days after completion of the work.

EXTERNAL INSPECTION

Time of Inspection.—The exterior of every boiler shall be thoroughly inspected before the boiler is put into service, and whenever the jacket and the lagging are removed.

Lagging to be Removed.—The jacket and the lagging shall be removed once every five years to permit inspection of the exterior of the boiler, or whenever the firebox is renewed, or whenever the Federal, State or railroad company's inspectors consider it desirable or necessary to thoroughly inspect the boiler.

HYDROSTATIC TEST

Pressure for Test.—All locomotive boilers must be subjected to a hydrostatic pressure of 25 percent above their rated working pressure, before being placed in service.

Frequency of Test.—This test must be made not less than once each twelve months.

Maximum Test Period.—Hydrostatic test will be due after twelve month's service, provided such service is performed within twenty-four consecutive months. For example: If a boiler receives a hydrostatic test and is placed in service for a period of six months and then placed in storage for fourteen months, and then again placed in service, it would be due for the next hydrostatic test in four months, at which time the twenty-four months' period has expired; in other words, the maximum period between hydrostatic tests for a boiler placed in service is twenty-four months, provided the boiler has not had more than twelve months' actual service.

Preparation for Test.—The water must be heated to about the boiling point immediately before pressure is applied.

Removal of Dome Cap.—The dome cap and throttle stand pipe must be removed at the time of making the hydrostatic test and the interior surface and connections of the boiler examined as thoroughly as conditions will permit. In case the boiler can be entered and thoroughly inspected without removing the throttle stand pipe, the inspector may make the inspection by removing the dome cap only, but the variation from this rule must be noted in the report of inspection.

Witness of Test.—When the test is being made, an authorized representative who is thoroughly familiar with boiler construction must personally witness the test and thoroughly examine the boiler while under test.

Reporting and Recording of Test.—A report of each hydrostatic test must be made on Forms 1 or 3, as well as record being made on Form No. 5.

Repairs and Steam Test.—When all necessary repairs have been completed, the boiler shall be fired up and a steam pressure raised to not less than the allowed working pressure and carefully examined for all defects in the boiler, seams, rivets, bolts, cocks, valves and appurtenances, which must be remedied before placing boiler in service.

STAYBOLT TEST

Frequency of Test.—All rigid staybolts of locomotives in service must be tested not less frequently than once each fifteen days. Staybolts must also be tested immediately after each hydrostatic test.

Preparation for Test.—When these tests are made, there must be not less than 50 pounds of steam pressure on the boiler, which will produce sufficient strain upon the staybolts to cause separation of the parts of broken ones. If the boiler is not under steam the examination may be made after drawing all the water from the boiler.

Method of Testing Rigid Staybolts.—An inspector especially trained for the service must tap with a hammer each staybolt and crown sheet stay accessible on the firebox end, and those not accessible must be tapped whenever possible on the outside end and judge if any are broken. Staybolts having telltale holes must in addition to the hammer test be carefully tested, to insure that all of the telltale holes are open, using a drill for this purpose when necessary.

Method and Frequency of Testing Flexible Staybolts.—All flexible staybolts having caps over the outer ends shall have the caps removed at least once every eighteen months, or whenever the United States or railroad company's inspector considers the removal desirable, in order to thoroughly inspect the staybolt. The firebox sheets to be examined carefully at least once a month, to detect any bulging or indications of broken staybolts.

All flexible staybolts without caps shall be tested once each month in the same manner as rigid bolts, and in addition shall be tested each time the hydrostatic test is applied while the boiler is under a hydrostatic pressure not less than the allowed steam pressure, proper notation of such test being made on Forms No. 1 or 3.

Removal of Caps—Time Limit.—Removal of caps from flexible staybolts will be due after eighteen months' service, providing such service is performed within thirty consecutive months. (Same explanation as covered under hydrostatic test applies to this rule also.)

Broken Staybolts.—No boiler shall be allowed to remain in service when there are one or more staybolts broken in the top row of firebox, nor two or more adjacent broken or plugged staybolts in any part of the firebox, nor three or more broken or plugged staybolts in a circle 4 feet in diameter, nor when five or more staybolts are broken in the entire boiler.

Removal of Staybolts.—Precaution must be taken to insure the removal of all defective staybolts, and a careful examination made of bolts adjacent to broken ones.

Telltale Holes.—All telltale holes and staybolts must be thoroughly cleaned by drill before hydrostatic test is ap-